

REVIEW ARTICLE

A Comprehensive Review of ConvNeXt Architecture in Image Classification: Performance, Applications, and Prospects

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ABSTRACT - The convergence of ConvNeXt architecture and classification tasks highlights an auspicious direction in modern computer vision. This review systematically analyzes how ConvNeXt transforms these challenges by examining its implementation from 2020 to 2025. This paper investigates four primary domains where ConvNeXt has demonstrated remarkable potential: medical imaging and diagnosis, general computer vision, fine-level image classification, and defect inspection. Analysis reveals significant performance improvements resulting from these ConvNeXt applications: Data augmentation methods consistently improve Bleu-1 detection metrics and F1-score on scarce datasets. SE (Squeeze-and-Excitation), CA (Channel Attention), and CBAM (Convolutional Block Attention Module) in the ConvNeXt model result in improved performance. ConvNeXt has produced outstanding performance, achieving 99.5% accuracy in medical imaging and brain tumor diagnosis classification. Collectively, these findings demonstrate how ConvNeXt, integrated with modern classification capabilities, continues to advance rapidly. Despite this progress, the ongoing challenge of training stability remains. By critically analyzing these advances and limitations, this paper provides important insights into the current state and potential future development of ConvNext-based classification systems.

Keywords: ConvNext Classification, Image, Computer Vision, Systematically Analyzes

1. Introduction

Computer vision is a rapidly growing field of research in computer science that focuses on enabling computers to understand, and interpret information from digital images and videos, similar to the human visual system. Significant advances in computer vision technology have been achieved thanks to innovations in more sophisticated architectures and training techniques. Convolutional Neural Networks (CNNs) have become the foundation of deep learning for processing grid-like data such as images, showing remarkable progress in various computer vision tasks such as feature extraction and classification. Furthermore, the advent of the Vision Transformer (ViT) has brought new perspectives to the field of computer vision with its outstanding performance in image classification and ability to learn long-range spatial relationships. Hybrid models, such as the ConvNeXt family, further enhance these capabilities by combining important features of both the ResNet and Vision Transformer architectures, resulting in better performance at lower computational cost. Various attention mechanisms, such as Coordinate Attention and Squeeze-and-Excitation Networks have also been introduced to further enhance the feature extraction capability and focus on important information in images. Computer vision applications are diverse, including image classification (e.g., glitch classification, hardware Trojan detection, skin disease classification, and face recognition), image segmentation (e.g., segmentation of medical images and remote sensing imagery), object detection, autonomous generation of medical reports from images, and visual tracking.

Deep learning is a subfield of machine learning that relies on artificial neural networks with many intermediate layers, where each layer builds on the representations learned by the previous layer to automatically develop increasingly complex and useful representations of the data, eliminating the need for manual feature engineering inherent in shallow learning methods. In the realm of computer vision, deep learning has made significant progress, driven by innovative architectures such as Convolutional Neural Networks (CNNs), which have strong generalization capabilities and self-learning capabilities to extract feature information from multiple levels in training images automatically. The ResNet family of architectures is a type of CNN that uses residual connections to enable deeper model training, while the Vision Transformer (ViT) has brought a new perspective by adopting a purely self-attention-based model that learns long-range spatial relationships. Hybrid models such as the ConvNeXt family further enhance these capabilities by combining key features of traditional CNN architectures, such as ResNet, with design principles inspired by the Vision Transformer, offering competitive performance at lower computational cost and improved feature extraction capabilities. Deep learning can be performed either supervised or self-supervised and has been widely applied in a variety of computer vision tasks, including image classification (e.g., glitch classification, hardware Trojan detection, skin disease classification, and face

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recognition), image segmentation (e.g., medical imagery and remote sensing), object detection, autonomous medical report generation from images, and visual tracking.

ConvNeXt is a deep learning architecture introduced by Meta AI Research in 2022, designed to demonstrate that traditional convolutional neural networks (CNNs) can achieve very competitive performance, if not outperform, the latest Vision Transformer (ViT) models. Based on the ResNet-50 architecture, ConvNeXt adopts modern design principles and training techniques inspired by the Vision Transformer (ViT) and Swin Transformer, such as the use of large kernels (e.g., 7x7 depthwise convolutions for non-local feature extraction capabilities similar to multi-head self-attention), reverse bottleneck structure, replacing ReLU activation functions with GELU, and using Layer Normalization (LN) instead of BatchNorm. ConvNeXt's strengths lie in its ability to achieve faster inference and higher accuracy compared to popular Transformer-based models, while maintaining lower computational cost and superior feature extraction capabilities. ConvNeXt efficiently captures long-range dependencies and hierarchical spatial features using separable depthwise convolutions. This architecture has been proven to be highly effective and robust in a variety of computer vision applications, including image classification (e.g. glitch classification, hardware Trojan detection, skin disease classification, mango leaf disease classification, aurora classification, and brain tumor grade classification) and image segmentation (e.g. remote sensing imagery, medical imagery), object detection (e.g. wall crack detection, ship target detection), autonomous medical report generation, visual tracking, text recognition, low-voltage equipment assembly quality inspection, mathematical expression search, and medical diagnosis.

Although the research is extensive and has tested the classifiers and ConvNeXt architecture independently, a comprehensive analysis that performs a comparison is needed. This review discusses the differences of this research systematically testing and comparing ConvNeXt in performing classification reported between 2020 and 2025. This analysis focuses on three domains in which ConvNeXt architectures address fundamental challenges: medical imaging and diagnosis, general computer vision, fine-level image classification, defect inspection, and classification. In each domain, this review compares the methodologies and performance evaluation frameworks reported by various researchers.

The contributions of this review are as follows:

- This article explores the use of ConvNeXt for classification tasks conducted during the period from 2020 to 2025.
- This study investigates the implementation of ConvNeXt in classification problems, organizing them according to their function in addressing key domain-specific issues, including medical image analysis, general vision tasks, detailed image classification, and defect detection. It also explains the specific architectural components of ConvNeXt employed in these applications.
- The paper offers a performance comparison of ConvNeXt-based solutions for classification, emphasizing how different techniques perform across a range of application areas.
- This review outlines the primary obstacles encountered and suggests potential future advancements to enhance classification capabilities with ConvNeXt.

2. Method

The application of deep learning in classification has received significant attention from the research community, resulting in a large number of survey papers examining various aspects of this field. While these surveys have provided valuable insights into classification techniques that deserve special attention, this section discusses relevant surveys to contextualize the current work and highlight unique contributions, as summarized in [Table 1](#).

Table 1. Mean accuracy values with and without Regularization for machine learning models (holdout validation)

Reference	Year	Primary focus	ConvNeXt Coverage	Main Contribution
[1]	2023	Classification of skin diseases	Extensive	Fusion-based CNN analysis for skin disease classification
[4]	2024	Medical image segmentation for disease detection and treatment	Moderate	Two-branch hybrid network (GLFUnet) analysis for medical image segmentation
[5]	2024	Brain tumor grade classification	Extensive	Brain Tumor Grade Classification Analysis
[13]	2024	Detection of diabetic macular edema (DME) biomarkers using OCT images	Moderate	Analysis, because it evaluates the efficiency of ConvNeXt and EfficientNet architectures for DME biomarker detection in OCT images.
[25]	2023	Diagnosis of Autism Spectrum Disorder (ASD) using sMRI and fMRI	Moderate	Analysis, investigating the use of vision transformer and 3D-CNN with a multi-slice generation technique for ASD diagnosis
[35]	2023	Classification of aurora structures	Extensive	Analysis, as it provides a direct comparison of deep learning tools for aurora structure classification.
[59]	2023	Detection of mango leaf disease	Extensive	Analysis of Mango Leaf Disease Detection System Based on ConvNeXt
[72]	2022	Crack detection algorithm in walls	Limited	CL-YOLOv8 analysis for crack detection with improved accuracy

Between 2020 and 2025, the application of deep learning in image classification has attracted significant attention from the research community. This has resulted in numerous review papers exploring various techniques and models used in classification tasks. One model that has shown significant progress is ConvNeXt, which utilizes modern convolutional neural network (CNN) architectures for image classification tasks, including more in-depth tasks such as aurora classification, plant disease detection, and medical diagnosis. In a study on aurora image classification, ConvNeXt demonstrated excellent results, with a peak accuracy of 98.5% in identifying aurora structures, demonstrating its ability to handle a wide range of complex image features [30]. ConvNeXt's application has also been demonstrated in plant disease detection systems, for example, in a case study on detecting mango leaf diseases caused by pathogens and pests. In this study, ConvNeXtXLarge demonstrated a peak accuracy of 100% for pathogen disease detection and 98.79% for pest detection in mango plants [23].

Furthermore, research has shown that the use of ConvNeXt in remote sensing image segmentation provides more accurate results compared to other models such as ResNet and Swin Transformer. This is due to ConvNeXt's ability to capture multi-scale context and better spatial features through the integration of innovative modules such as Global Pyramid Guidance (GPG), Residual Atrous Spatial Pyramid Pooling (RASPP), and Scale-Aware Pyramid Fusion (SAPF). These modules enable ConvNeXt to address the challenges of distinguishing objects of the same material or objects occluded by shadows [19],[20]. The application of ConvNeXt has also demonstrated outstanding results in disease detection tasks in medical images, such as brain tumor classification using the BraTS 2019 dataset. The results of this study showed that using ConvNeXt pre-trained on a common dataset like ImageNet, the model can achieve up to 99.5% accuracy in brain tumor classification, demonstrating its ability to process and classify limited medical data [17].

Overall, these studies demonstrate that ConvNeXt is a highly effective and efficient model, particularly in addressing the limitations of small datasets and complex medical or agricultural image processing. This model not only improves detection accuracy but also speeds up the training process and reduces computational costs, which are crucial in real-world applications such as agriculture and medical diagnosis.

3. Results and Discussion.

ConvNeXt has demonstrated remarkable performance across various classification tasks, particularly in the classification of aurora images. Achieving an impressive 98.5% accuracy, the model surpasses others like ResNet-50, MobileViT, and Swin Transformer. This highlights ConvNeXt's strong capability in processing intricate image features. It consistently outperforms competing models in precision, recall, and F1 score, which emphasizes its effectiveness in classification tasks, particularly when distinguishing challenging aurora features such as diffuse and discrete auroras [1]. The model excels in handling complex images and delivers accurate results, even when faced with challenging data. For instance, in the task of corn disease classification, ConvNeXt achieved 95.2% accuracy, outperforming models such as ResNet34 and MobileNetV2 [6]. Its ability to process image data efficiently makes it a preferred choice in tasks involving both medical and complex imagery [2].

A key strength of ConvNeXt lies in its ability to manage diverse image features without significant loss of performance. In aurora classification, ConvNeXt identifies various aurora classes with high precision and recall [5]. This is crucial because many aurora classes share similar shapes, making it essential for the model to capture subtle differences between them [1]. Furthermore, ConvNeXt stands out in transfer learning, which enhances accuracy and minimizes overfitting risks. When trained using pre-labeled aurora data, the results are significantly better compared to training from scratch. The integration of the SimAM attention module further boosts the model's capacity to focus on critical features, improving overall performance across multiple datasets [2][1]. Despite its strong performance, ConvNeXt still offers potential for further enhancement. The adoption of more refined data augmentation techniques and model architecture improvements could make the model even more adaptable to varying data conditions. This could include better handling of imbalanced data or fluctuating lighting conditions, which sometimes impact prediction accuracy [2][3].

In conclusion, ConvNeXt has proven itself to be a robust and efficient model for image classification tasks. Its superior handling of complex data, coupled with its ability to leverage transfer learning, makes it an excellent choice for various applications in both computer science and the medical field. This model is highly suitable for image classification systems where high accuracy is critical, such as in disease detection.

Table 2. Performance summary of ConvNext methods for Classification

Ref	Method	Dataset	Result		Drawbacks
			mAP (%)	F1-score (%)	
[1]	ConvNeXt	HAM10000.	-	89.99%.	The complexity of the model increases due to the fusion technique.
[2]	YOLOv8n + ConvNeXt V2 + LSKA	Brickwork Cracks Dataset	83.7%	-	Limitations on crack detection in extreme lighting conditions.
[3]	3D-CNN + ConvNeXt, ViT, Swin, MobileNet	ABIDE-NYU subset + sMRI & fMRI	-	84.6%	Performance is highly dependent on slice quality and preprocessing.

Ref	Method	Dataset	Result		Drawbacks
			mAP (%)	F1-score (%)	
[4]	ResNet18, ResNet34, ConvNeXt	Gravity Spy v1.0	-	97.18%	Problems persist with minority glitches (such as Paired Doves, which only has 27 samples)
[5]	ResNet-50 + MobileViT-XXS + Swim-T + ConvNeXt-Tiny	THEMIS All-Sky Imager		99.6%	The model was tested only on the THEMIS dataset; it has not been tested on other aurora domains.
[6]	ConvNeXt + SimAM attention module dan transfer learning	8 kelas penyakit jagung dengan 17,670 gambar Dataset visual	-	95%	Only tested on a single augmentation dataset, not on real field data
[7]	YOLOv5s + MobileNetV3-Small + CNeB (ConvNeXt-based Block)	inspeksi kapal (real-world), tersedia di GitHub	97.5%	-	Focus on lightweight, less in-depth exploration of small-target detection

ConvNeXt-based models have shown notable progress in managing occlusions across various tasks. In pedestrian detection, ConvNeXt surpassed models like AlexNet by extracting more detailed and reliable features from input data, which improved its accuracy in detecting distracted pedestrians, even when occlusions or distractions occurred. This ability to handle occlusions effectively is due to ConvNeXt's deep learning strengths, enabling it to capture both high-level and intricate features [15]. In the realm of remote sensing image segmentation, ConvNeXt-powered models like CMPF-UNet utilize multi-scale pyramid fusion to boost feature extraction, resulting in enhanced performance when dealing with occluded regions. By combining contextual information at various scales, ConvNeXt was able to manage complex scenarios, including occluded objects, better and maintain a clear understanding of the overall scene. This was demonstrated in challenging datasets such as Vaihingen and Potsdam, where occlusions often hid parts of the image [20].

For visual tracking, ConvNeXt's deep feature extraction, coupled with an anchor-free design, proved to be highly effective in ensuring the model remained robust even in occluded scenarios. The model's focus on learning rich representations without relying on predefined anchor boxes made it adaptable to complex tracking situations involving occlusions [8]. Furthermore, ConvNeXt's performance in RGB-D semantic segmentation illustrated its ability to handle occlusions effectively. By integrating both RGB and depth data, ConvNeXt-based models improved segmentation accuracy, even when parts of the objects were hidden. The fusion of these modalities helped the model overcome occlusion challenges and achieve high accuracy in complex or cluttered scenes [9].

Table 3. Performance summary of ConvNext methods for handling Occlusions in Classification

Ref	Method	Dataset	Result	Drawbacks
[8]	ConvNeXt backbone + 3D-CMax fusion + anchor-free Siamese tracking	OTB100, UAV123, GOT-10K	AUC (OTB100): 0.549. Precision: 0.756 FPS: 58.44 (real-time)	Real-time performance degrades slightly in environments with a lot of interference.
[9]	ConvNeXt + Cross-Modal Feature Fusion Module (CMFFM)	NYUDv2, SUN-RGBD, RICE- RGBD	- mIoU (RICE- RGBD): 74.8% - Pixel Accuracy: 88.3%	Performance is limited on datasets with extreme occlusion in high-density environments.
[11]	ConvNeXt pre-trained + Linear Probing	OTB100, UAV123, GOT-10K	Performance is limited on datasets with extreme occlusion in high-density environments	Performance degradation on full occlusion with motion blur
[15]	CAFSN dengan ConvNeXt backbone dan modul 3D-CMax Fusion	OTB100	AUC: increased from 0.525 (ResNet) to 0.549 Precision: 0.756	3D-CMax Fusion delivers better performance in handling occlusions.
[20]	ConvNeXt + Multi-Scale Pyramid Fusion (SAPF) + Residual Atrous Spatial Pyramid Pooling (RASPP)	Vaihingen, Potsdam	mIoU: 80.23% Overall Accuracy (OA): 89.17%	computational improvements on larger datasets and deeper models,

JujubeNet, which is based on ConvNeXt, was designed to perform highly accurate small object classification. By integrating a Multi-Branch Depthwise Separable Convolution (MDC) module and the Convolutional Block Attention Module (CBAM), JujubeNet achieved 99.1% accuracy on the 'Jujube2000' dataset. Additionally, the model reduces computational demands significantly, with its parameters and FLOPS being only 30% of ConvNeXt-Tiny, making it efficient for large-scale defect detection [35]. Similarly, MpoxNet, which also utilizes ConvNeXt, features a dual-branch design with a Squeeze and Excitation (SE) module to improve feature extraction. It achieved 95.28% accuracy while maintaining lower complexity, with parameters only 30.68% of ConvNeXt-Tiny. This efficient architecture is suitable for fast and accurate disease detection, especially in healthcare environments. SA-ConvNeXt combines ConvNeXt with the Selective Attention Network (SANet) and several attention mechanisms, including SimAM and M-ECA. This model demonstrated 96.7% accuracy and 98.2% recall on the Oxford Flowers102 dataset. By integrating multiple attention strategies, SA-ConvNeXt enhances feature extraction and outperforms traditional models such as VGG16 and ResNet. [40]

In conclusion, these ConvNeXt-based models showcase significant improvements in small object classification. They incorporate advanced attention mechanisms and refined structures to deliver high accuracy while keeping computational costs low. Their ability to process complex data with fewer parameters makes them highly effective for a range of applications, from industrial defect detection to medical diagnosis [31].

Table 4. Performance summary of ConvNext methods for small object Classification

Ref	Method	Performance	Drawbacks
[31]	ConvNeXt	Classification accuracy is 8.85% higher than the previous model.	Sensitivity to data with small variations between classes is low, requires additional data augmentation.
[32]	IWNeXt, ConvNeXt	PSNR increased by 1,651 dB compared to the previous model (DuDoRNet).	Limited to the MRI medical image domain, it does not cover other variations.
[33]	ConvNeXt + VGG16	99.1% accuracy, 99.1% recall, 99.1% precision	Less efficient with lighting variations or complex backgrounds image
[35]	MpoxNet berbasis ConvNeXt	Accuracy 95.28%, precision 96.40%, recall 93.00%	Not effective in detection with limited data or different video manipulation data
[40]	SA-ConvNeXt	Accuracy 96.7%, recall 98.2%	Requires high computational capacity and longer training time on datasets

3.1 Challenges and Future Directions for ConvNeXt in Classification

Although ConvNeXt performs well in image classification tasks, it still faces several challenges that require attention to fully unlock its potential in real-world applications. One major issue is optimizing the model for diverse and complex datasets. While ConvNeXt handles high-resolution images effectively, its performance can decline in extreme conditions, such as low-light environments, occlusions, or highly varied image features. This shows the need for further improvements in the model's robustness, especially in situations where visual clarity is affected. Furthermore, ConvNeXt's dependence on significant computational resources for training and fine-tuning creates a barrier for use in environments with limited resources. For example, real-time image processing tasks, like those in autonomous vehicles or edge devices, may find it difficult to utilize ConvNeXt without major hardware upgrades or optimization efforts. The adaptability of ConvNeXt to a broader range of tasks and domains also remains an ongoing challenge. While the model has shown great promise in specific domains like medical imaging and aurora classification, its application to other fields may require further adaptation of the architecture and training methodologies. This is especially relevant in specialized domains where labeled data may be scarce or domain-specific features need to be better captured. A major future direction for ConvNeXt involves the exploration of domain-adaptive techniques and transfer learning strategies that can enable the model to generalize better across different datasets, even in situations where training data is limited.

In terms of future research directions, improving the efficiency of ConvNeXt for deployment in real-time applications should be a key priority. This includes research into lightweight architectures and model compression techniques that reduce computational costs while maintaining high accuracy. In particular, hybrid approaches that integrate ConvNeXt with other architectures, such as transformers, could provide opportunities to enhance feature learning capabilities and achieve faster inference times. Additionally, the exploration of multi-task learning frameworks, where ConvNeXt can simultaneously handle both classification and enhancement tasks, holds promise for improving overall system efficiency. Another promising direction for future development is addressing the model's training stability and fine-tuning processes. Developing more advanced optimization algorithms, along with self-supervised or weakly supervised learning techniques, could help alleviate the need for extensive labeled datasets. Moreover, investigating adaptive training strategies that dynamically adjust to different environmental conditions could further enhance ConvNeXt's robustness. The ability to transfer learned knowledge across domains and tasks, through techniques like transfer learning, also has the potential to improve ConvNeXt's generalization capacity and enable better performance in diverse application areas.

Finally, the need for standardized evaluation metrics and benchmarking frameworks remains a key aspect of future research. Creating comprehensive evaluation systems that combine both classification accuracy and computational efficiency, along with a unified set of metrics for assessing model performance across different tasks and environmental conditions, would help establish more reliable comparisons and best practices. As these challenges are addressed, ConvNeXt's ability to handle complex and varied classification tasks will continue to evolve, potentially making it a standard in image recognition systems across diverse industries.

4. Conclusions

This research highlights the intersection between the ConvNeXt development model and image classification applications, highlighting the significant advancements and innovations expected between 2020 and 2025. During this

period, ConvNeXt has demonstrated remarkable progress in addressing complex image classification tasks, including improvements in accuracy, computational efficiency, and the ability to handle diverse image data types. With its ability to address challenges in domains such as aurora recognition and medical imagery, ConvNeXt has proven its superiority as a reliable model in real-world applications.

However, despite these remarkable achievements, challenges related to resource efficiency, training stability, and generalization across diverse conditions and domains remain significant obstacles that need to be addressed. Future research should focus on developing more lightweight architectures, transfer learning techniques, and adaptive training strategies that can address the limitations of labeled data. Thus, while ConvNeXt has achieved significant milestones in image classification, further research focused on improving the robustness, efficiency, and adaptability of this model will pave the way for broader and more efficient applications in various fields in the future.

Conflicts of Interest

The author declares no conflict of interest.

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